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import java.io.*;
import java.util.Scanner;

public class EmployeeManagementSystem {
    private static final String FILE_NAME = "employees.txt";

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        while (true) {
            System.out.println("\n--- Employee Management System ---");
            System.out.println("1. Add Employee Record");
            System.out.println("2. Display All Employee Records");
            System.out.println("3. Search Employee Record");
            System.out.println("4. Exit");
            System.out.print("Enter your choice: ");

            String choice = scanner.nextLine().trim();
            switch (choice) {
                case "1":
                    addEmployee(scanner);
                    break;
                case "2":
                    displayEmployees();
                    break;
                case "3":
                    searchEmployee(scanner);
                    break;
                case "4":
                    System.out.println("Exiting application. Goodbye!");
                    scanner.close();
                    System.exit(0);
                default:
                    System.out.println("Invalid option. Please choose between 1
and 4.");
            }
        }
    }

    /**
     * Case 1 & 2: Adds a new employee record using character streams.
     */
    private static void addEmployee(Scanner scanner) {
        try {
            System.out.print("Enter Employee ID: ");
            String id = scanner.nextLine().trim();
            if (id.isEmpty()) {
                System.out.println("Error: Employee ID cannot be empty.");
                return;
            }

            System.out.print("Enter Employee Name: ");
            String name = scanner.nextLine().trim();

            System.out.print("Enter Department: ");
            String department = scanner.nextLine().trim();

            System.out.print("Enter Salary: ");
            String salaryStr = scanner.nextLine().trim();

            // Validate salary formatting
            double salary;
            try {
                salary = Double.parseDouble(salaryStr);
            } catch (NumberFormatException e) {

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        System.out.println("Error: Invalid salary amount. Please enter a
valid number.");
        return;
    }

    // Using Try-With-Resources to automatically close character streams
    try (FileWriter fw = new FileWriter(FILE_NAME, true);
        BufferedWriter bw = new BufferedWriter(fw);
        PrintWriter out = new PrintWriter(bw)) {

        // Store record as a comma-separated line
        out.println(id + "," + name + "," + department + "," + salary);
        System.out.println("Employee record added successfully");
    }
} catch (IOException e) {
    System.out.println("An error occurred while writing to the file: " +
e.getMessage());
}

/**
 * Case 3: Displays all stored records from the text file.
 */
private static void displayEmployees() {
    File file = new File(FILE_NAME);

    // Custom check to see if the file exists and is populated
    if (!file.exists() || file.length() == 0) {
        System.out.println("Error: No employee records found. The file does
not exist or is empty.");
        return;
    }

    System.out.println("\n--- Stored Employee Records ---");
    System.out.printf("%-15s %-20s %-20s %-10s\n", "Employee ID", "Name",
"Department", "Salary");

    System.out.println("-----");

    try (FileReader fr = new FileReader(FILE_NAME);
        BufferedReader br = new BufferedReader(fr)) {

        String line;
        while ((line = br.readLine()) != null) {
            String[] details = line.split(",");
            if (details.length == 4) {
                System.out.printf("%-15s %-20s %-20s %-10s\n", details[0],
details[1], details[2], details[3]);
            }
        }
    } catch (IOException e) {
        System.out.println("An error occurred while reading the file: " +
e.getMessage());
    }

    /**
     * Optional Feature: Search an individual record using Employee ID.
     */
    private static void searchEmployee(Scanner scanner) {
        File file = new File(FILE_NAME);
        if (!file.exists() || file.length() == 0) {
            System.out.println("Error: Cannot search. The file does not exist or

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is empty.");
    return;
}

System.out.print("Enter Employee ID to search: ");
String searchId = scanner.nextLine().trim();
boolean found = false;

try (FileReader fr = new FileReader(FILE_NAME);
    BufferedReader br = new BufferedReader(fr)) {

    String line;
    while ((line = br.readLine()) != null) {
        String[] details = line.split(",");
        if (details.length == 4 &&
details[0].equalsIgnoreCase(searchId)) {
            System.out.println("\n--- Record Found ---");
            System.out.println("Employee ID: " + details[0]);
            System.out.println("Name: " + details[1]);
            System.out.println("Department: " + details[2]);
            System.out.println("Salary: " + details[3]);
            found = true;
            break;
        }
    }

    if (!found) {
        System.out.println("Message: Record with Employee ID '" +
searchId + "' not available.");
    }
} catch (IOException e) {
    System.out.println("An error occurred during search operations: " +
e.getMessage());
}
}
}

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